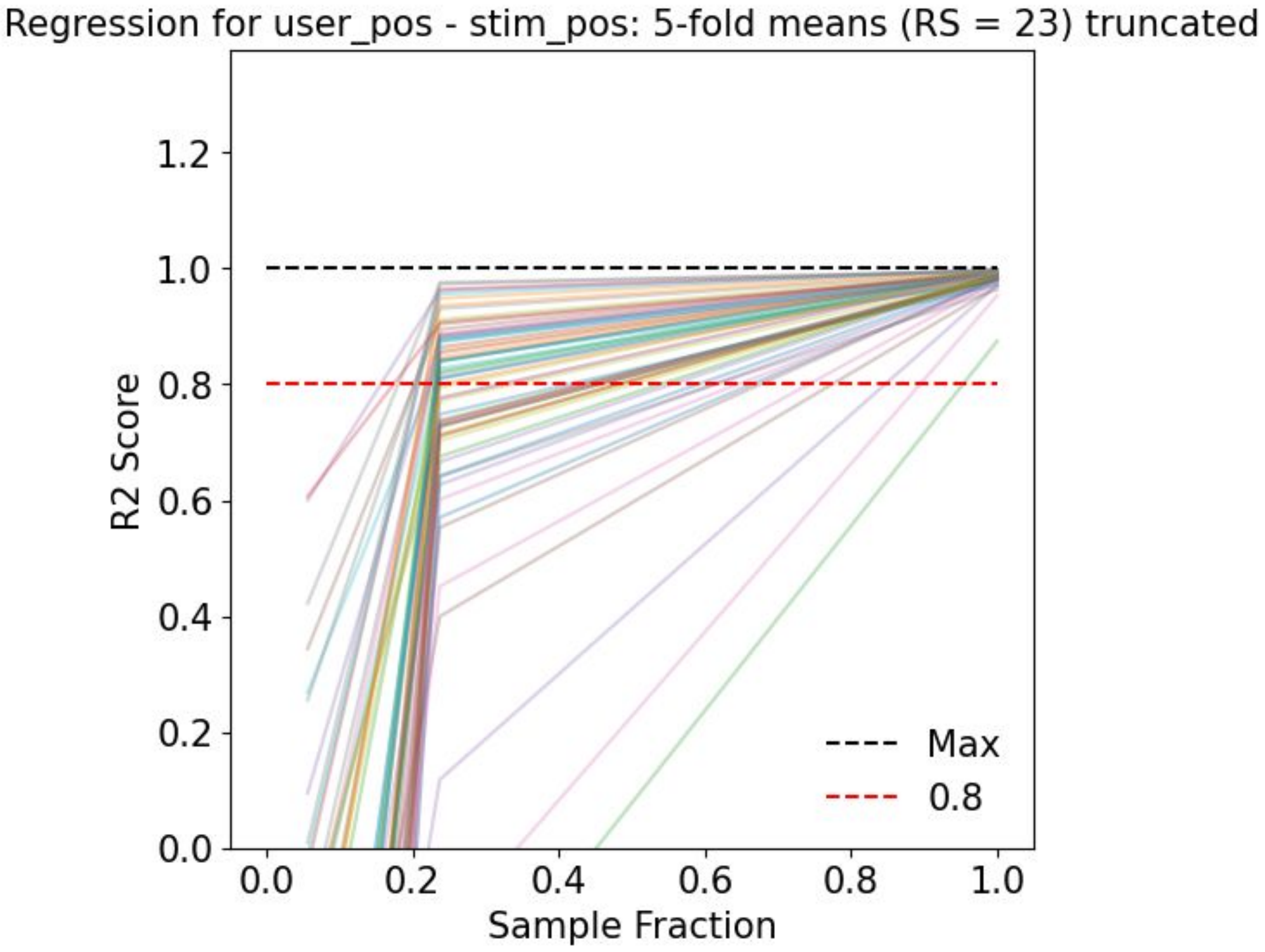
Team MoBI: Final Results

11/19/2024

Team Members: Liora Dsilva, Isha Dev, Taiming Chen, Deborah Roman, Sai Pranav Avva, Riya Modi

Results - Random Forest Regression

- Target: Position difference
- 5 fold cross validation
- 61 subjects, ~17000 raw data streams



Results - Random Forest Regression

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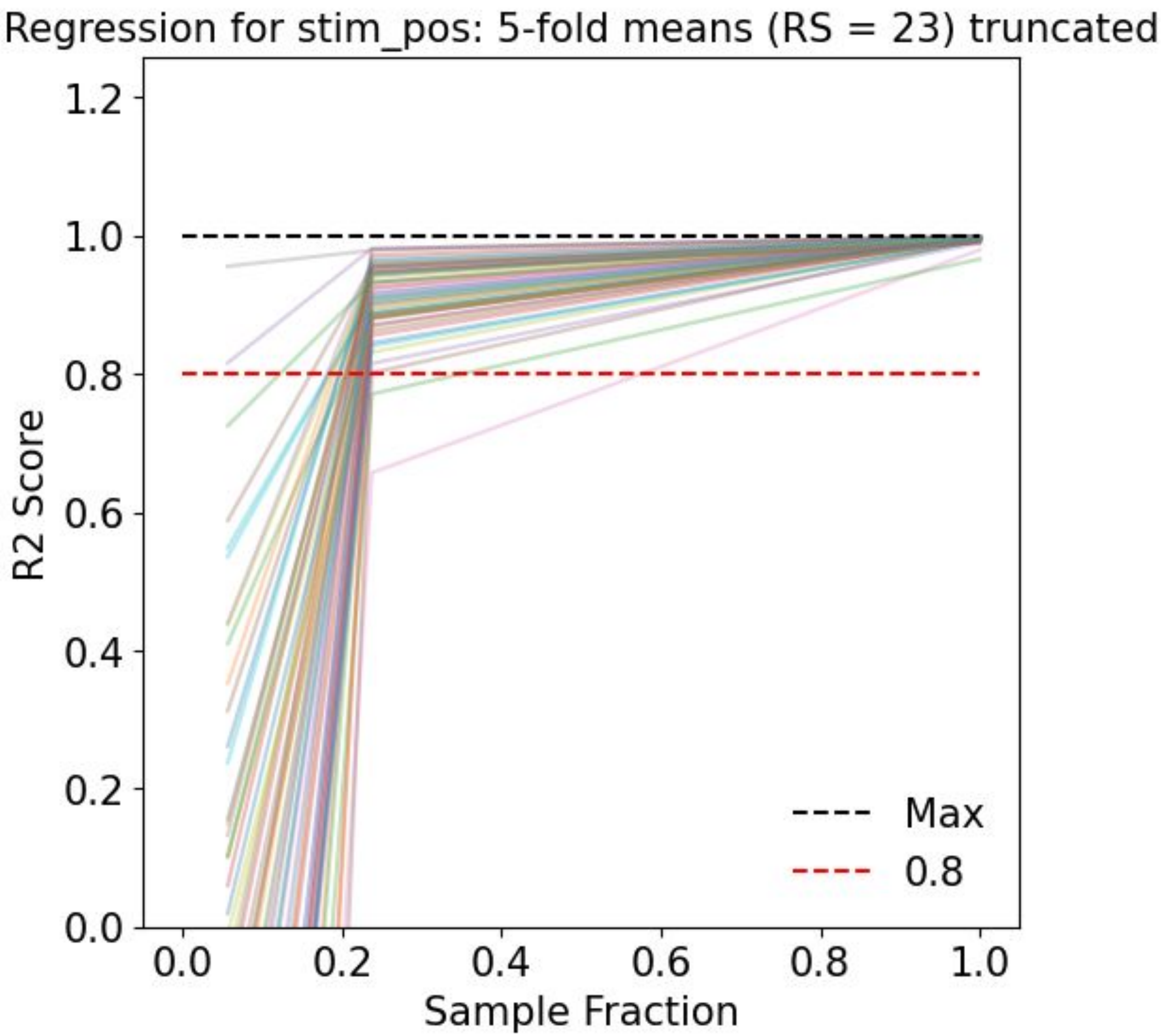
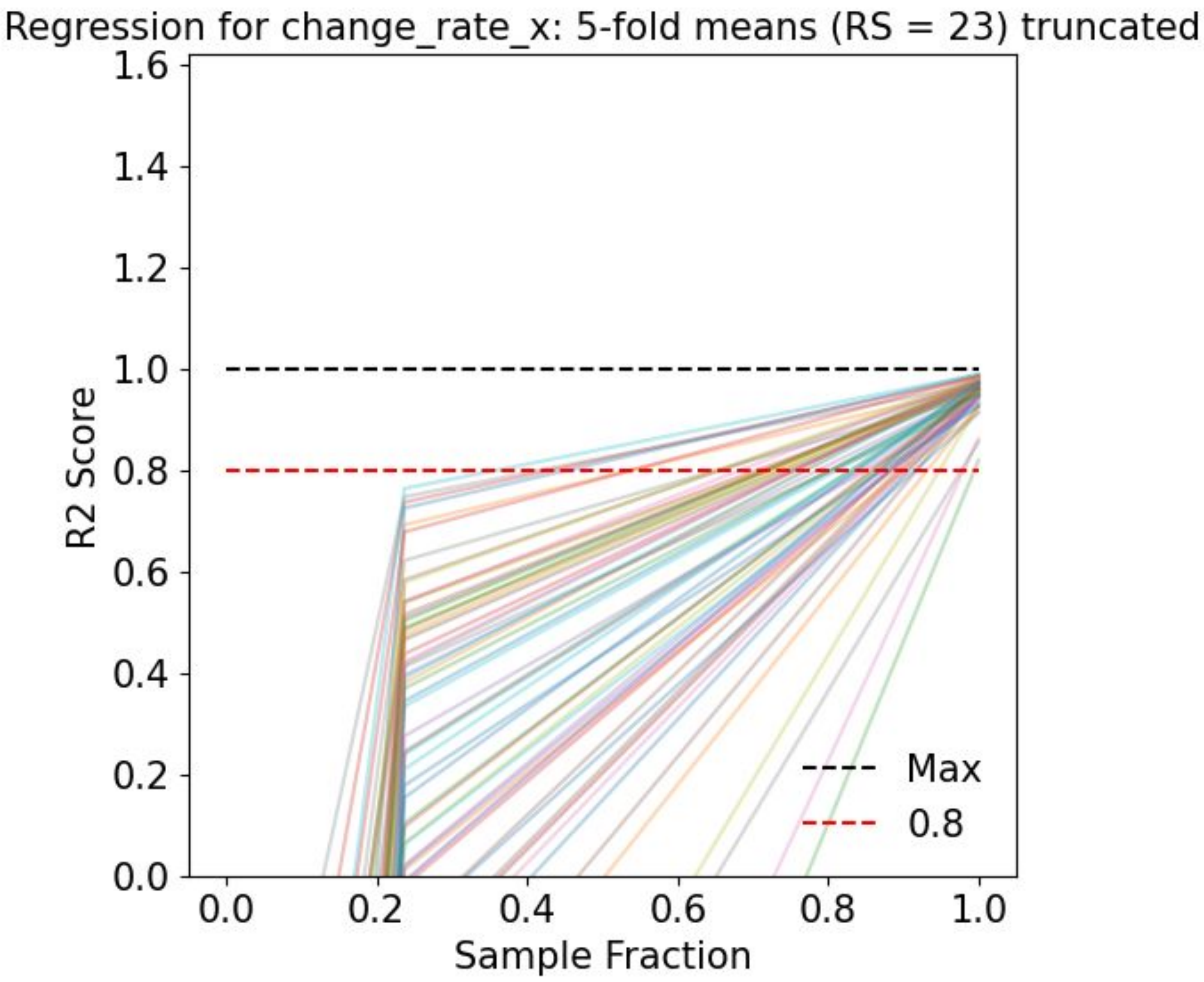
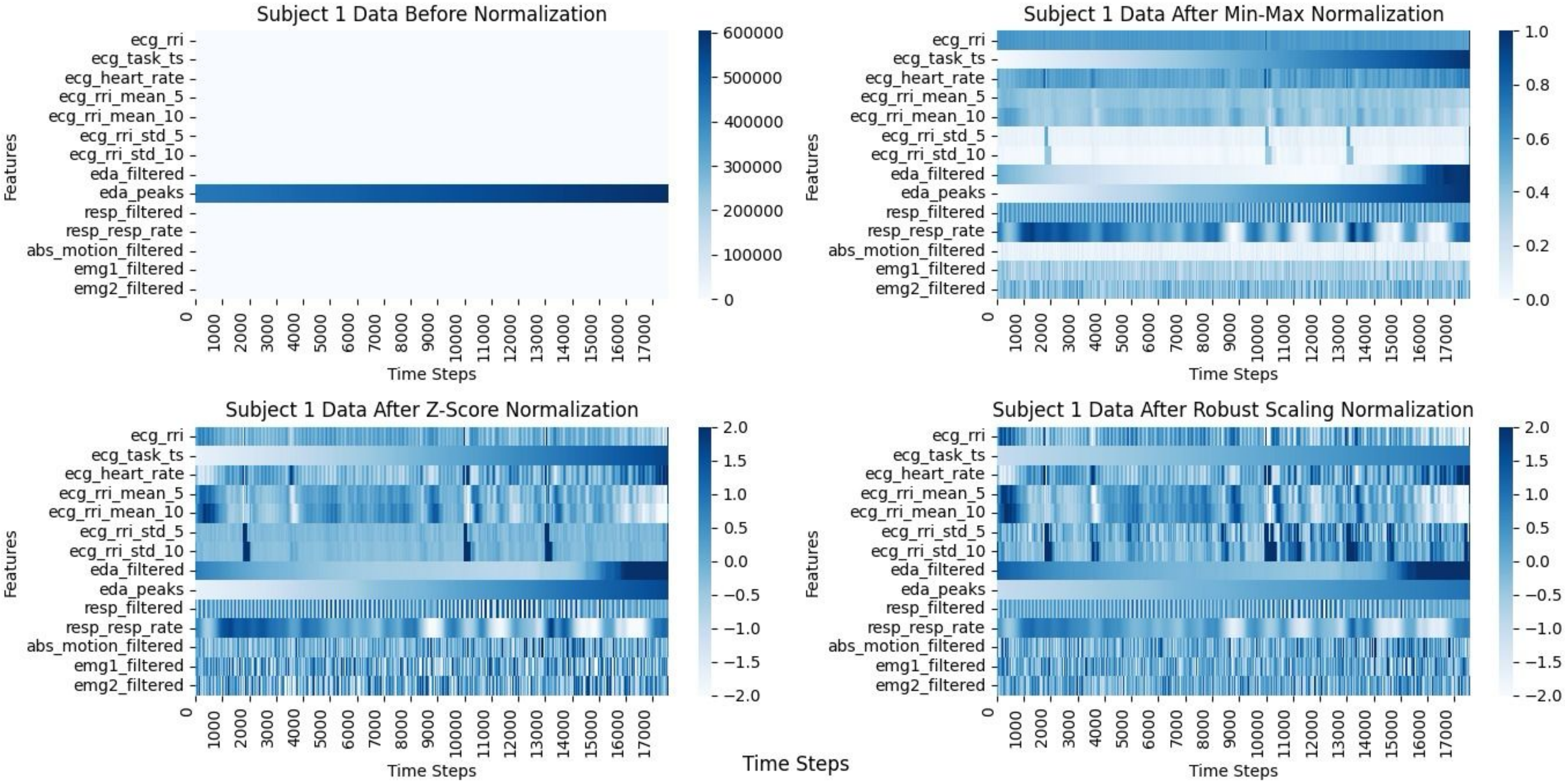


Results - Random Forest Regression

- Target: Stimulus speed
- 5 fold cross validation
- 61 subjects, ~17000 raw data streams



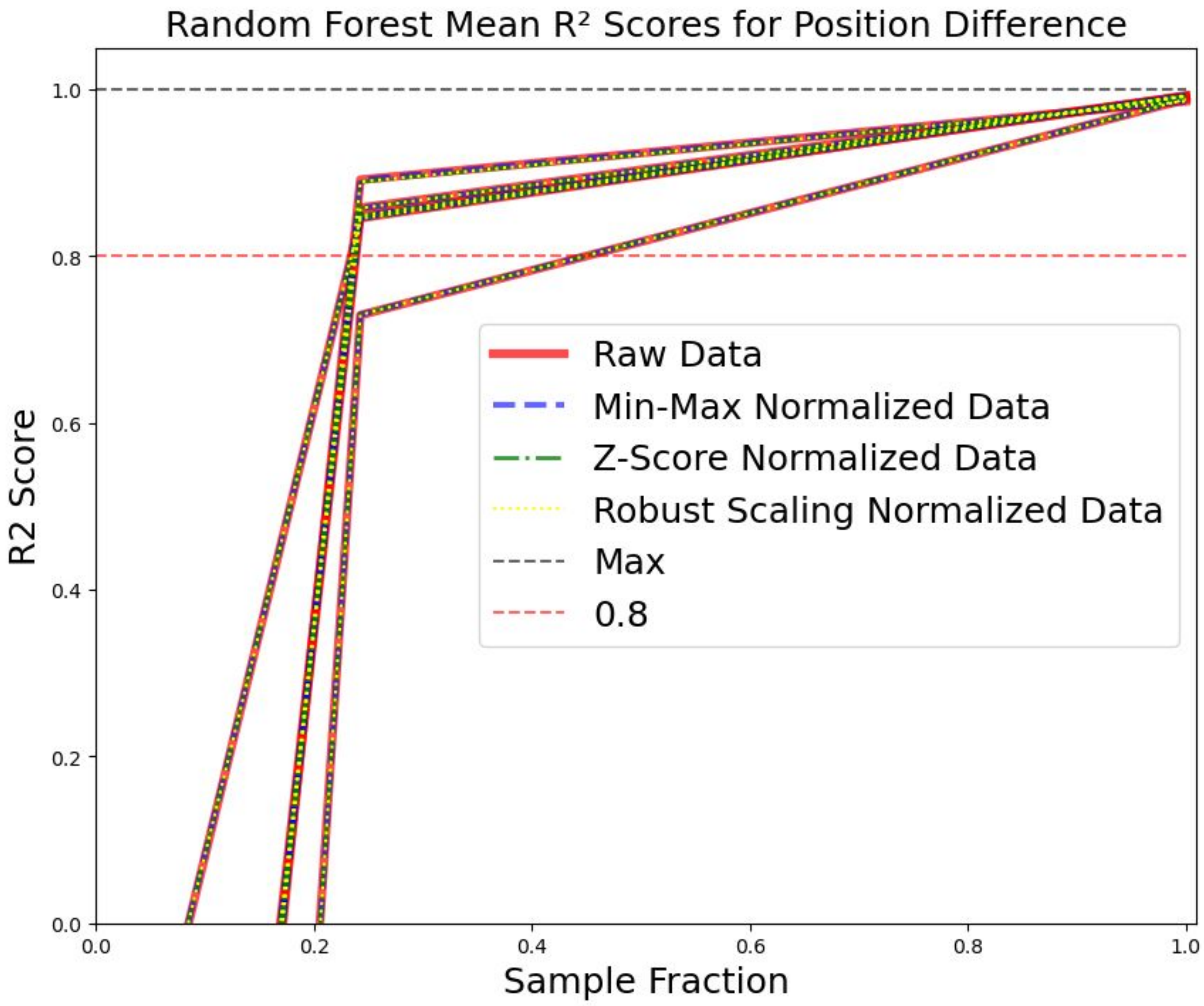
Results - Normalization



Time Steps

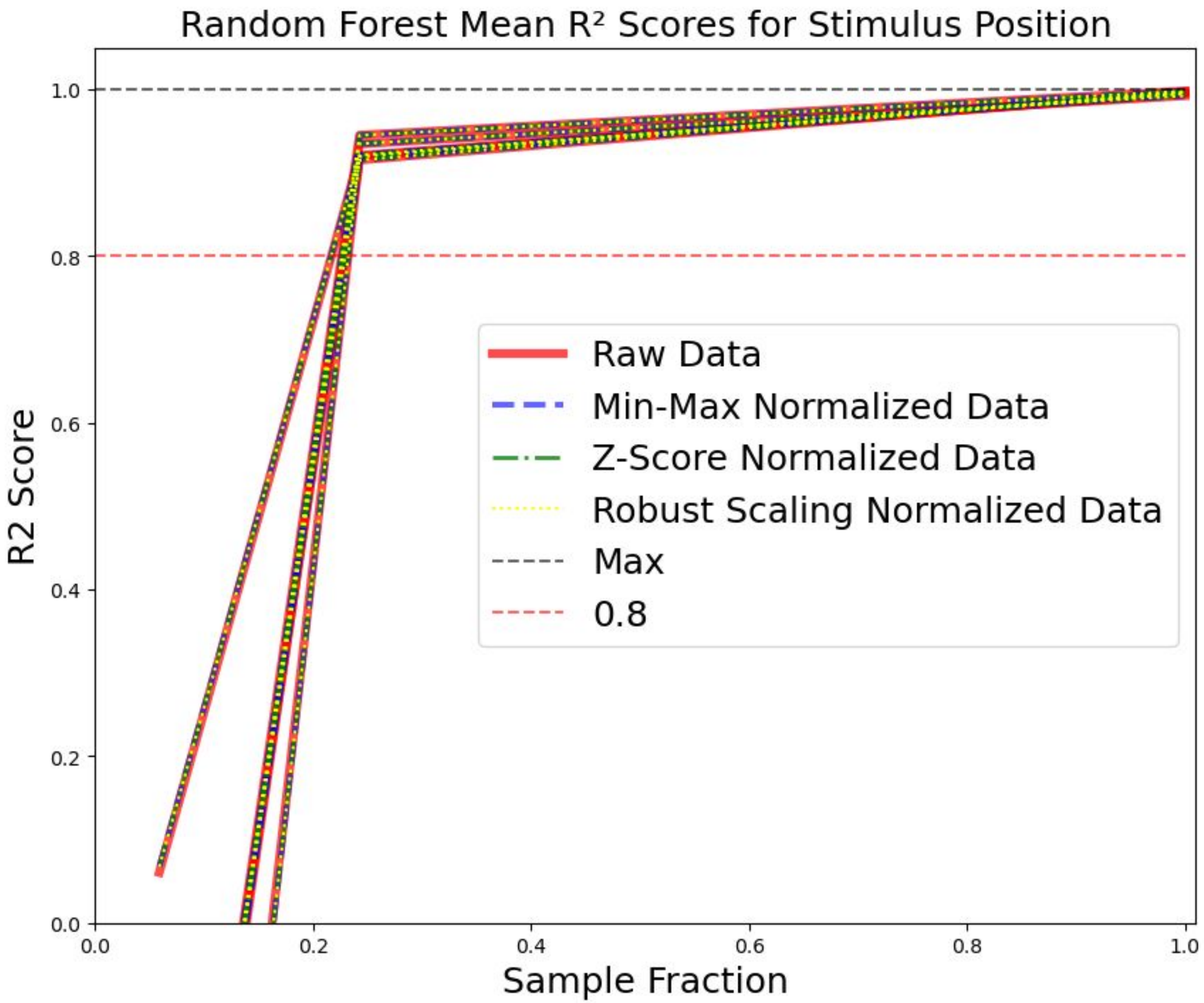
Results - Random Forest Regression

- Target: Position difference
- 5 fold cross validation
- 5 subjects, ~17000 normalized data streams



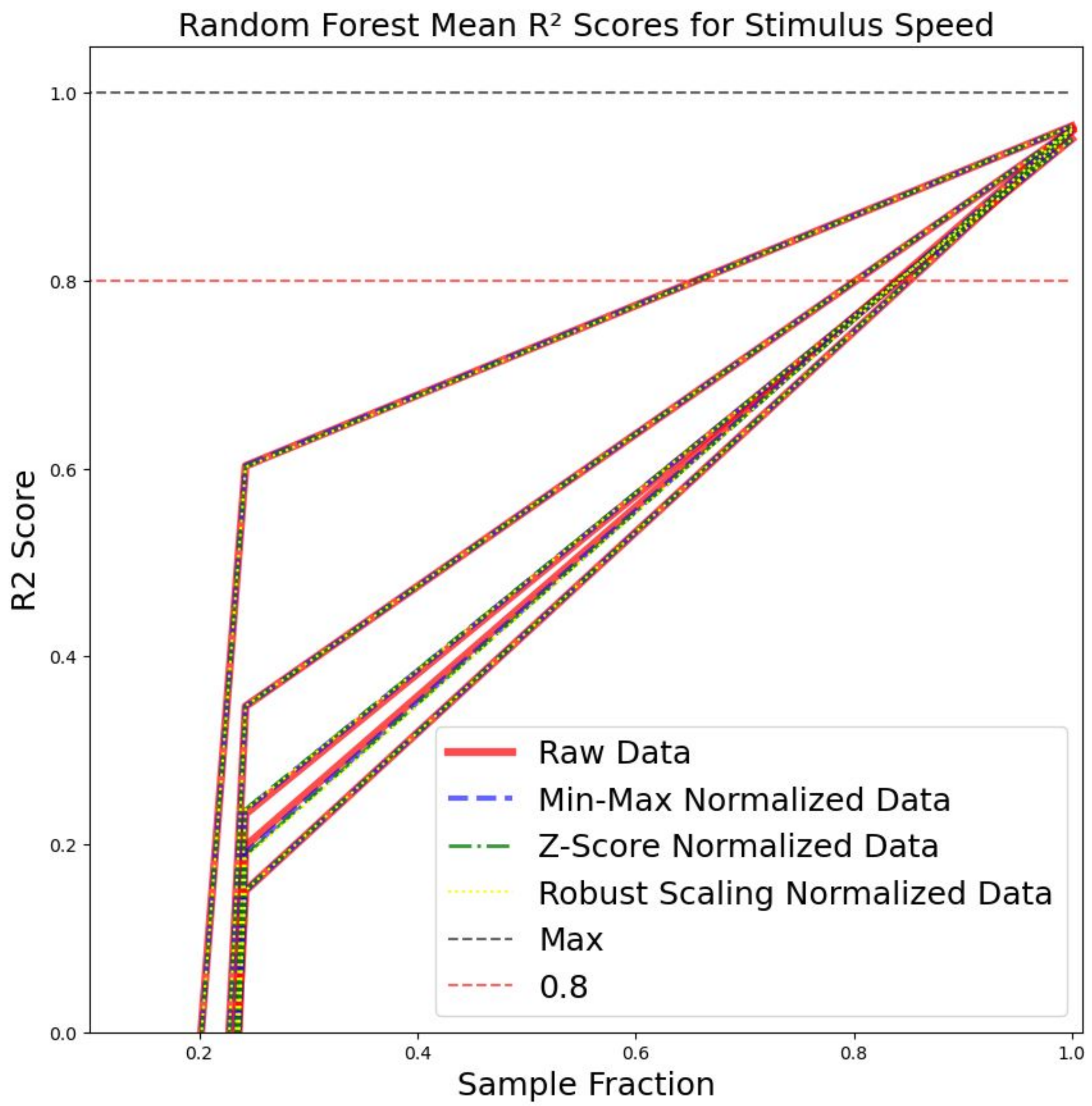
Results - Random Forest Regression

- Target: Stimulus position
- 5 fold cross validation
- 5 subjects, ~17000 normalized data streams



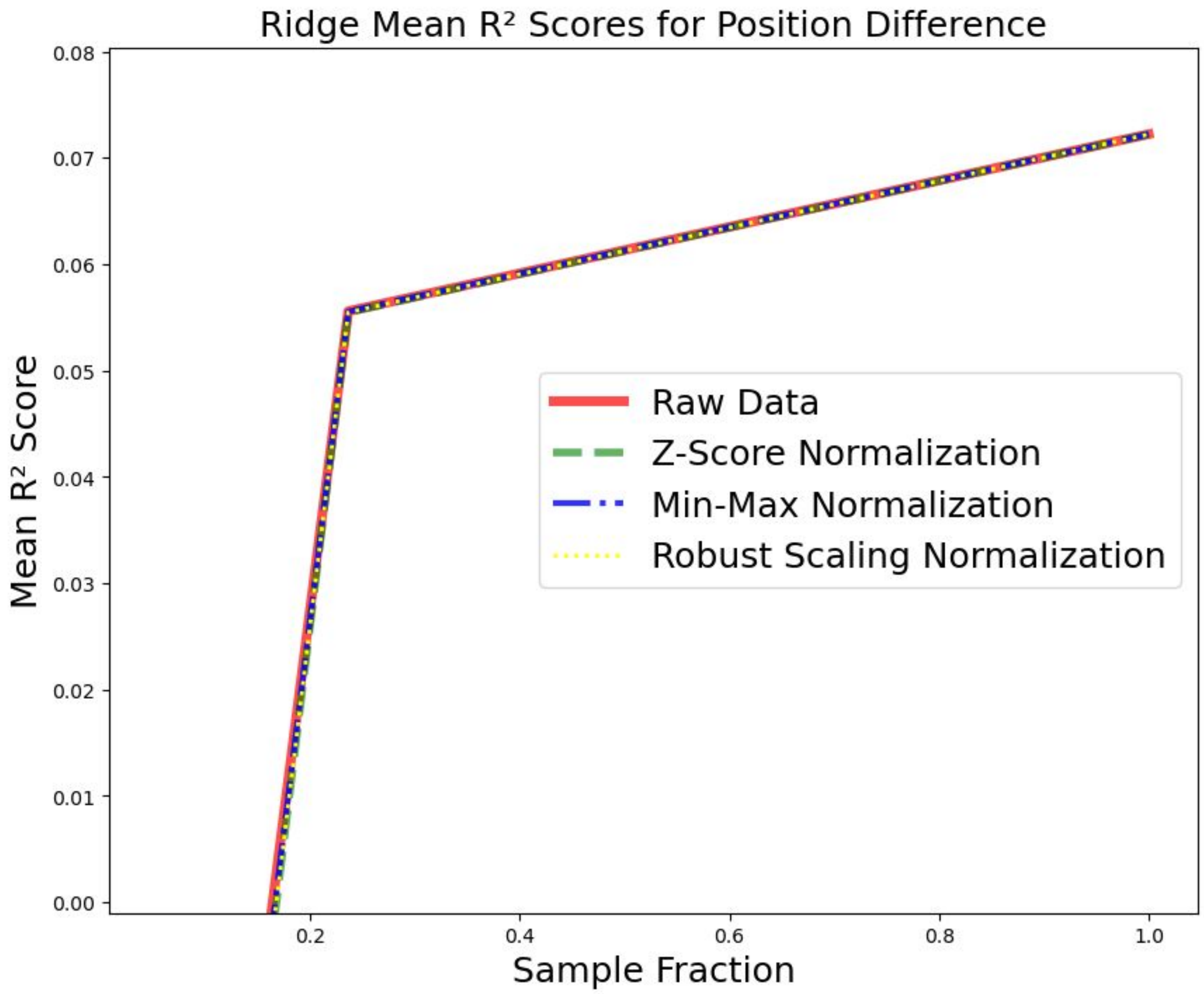
Results - Random Forest Regression

- Target: Stimulus speed
- 5 fold cross validation
- 5 subjects, ~17000 normalized data streams



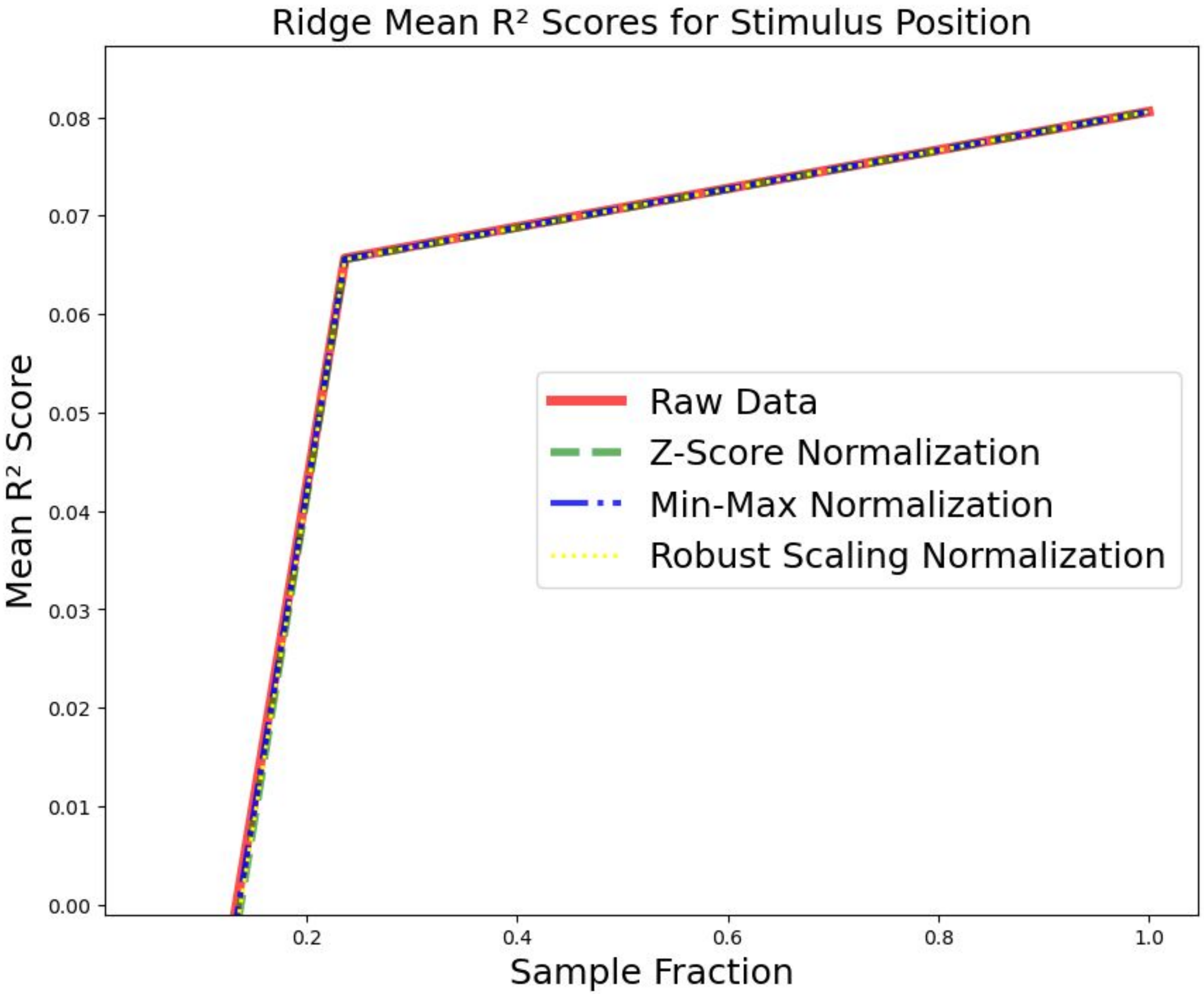
Results - Ridge Regression

- Target: Position difference
- 5 fold cross validation
- 1 subject, ~17000 normalized data streams



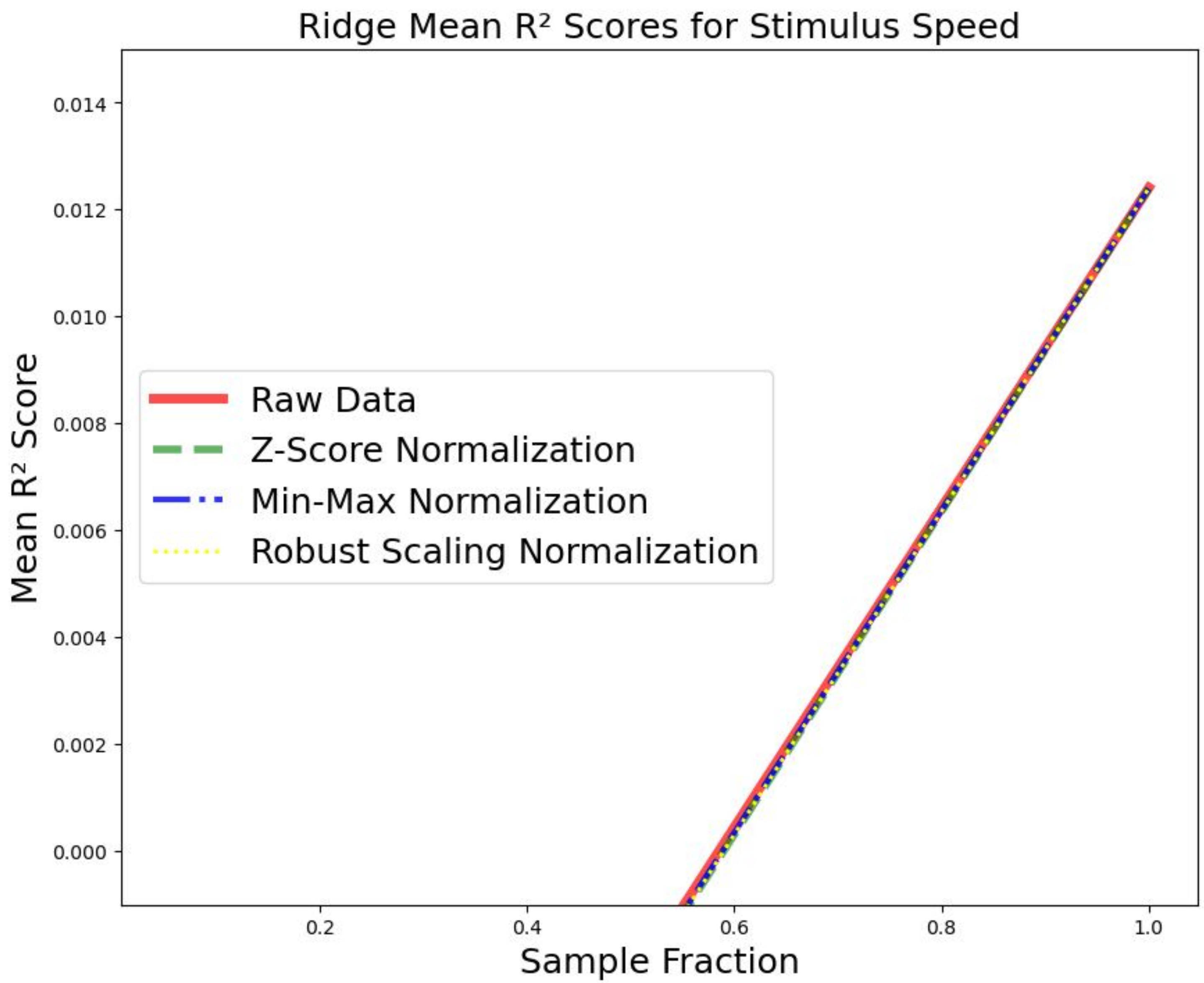
Results - Ridge Regression

- Target: Stimulus position
- 5 fold cross validation
- 1 subject, ~17000 normalized data streams



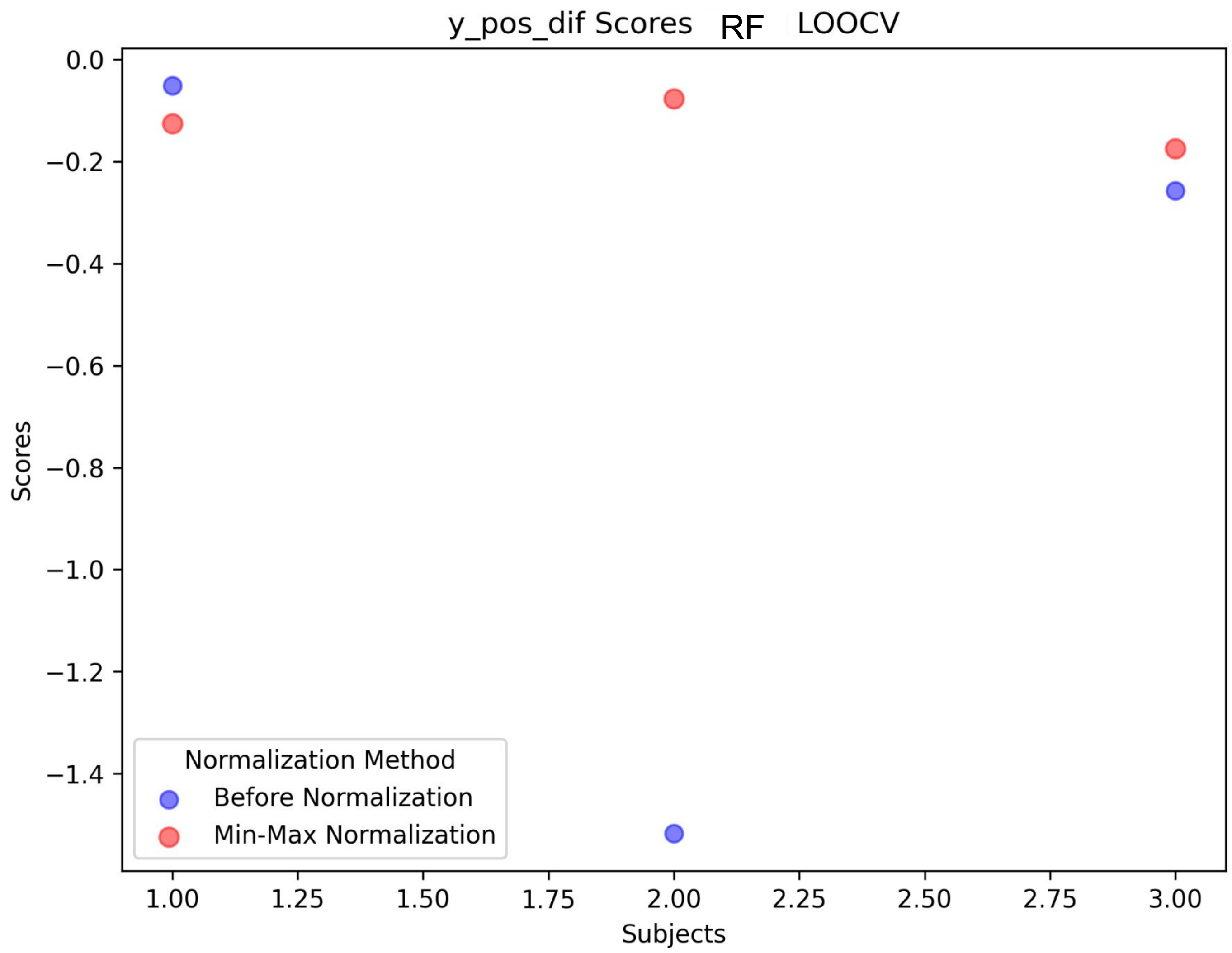
Results - Ridge Regression

- Target: Stimulus speed
- 5 fold cross validation
- 1 subject, ~17000 normalized data streams



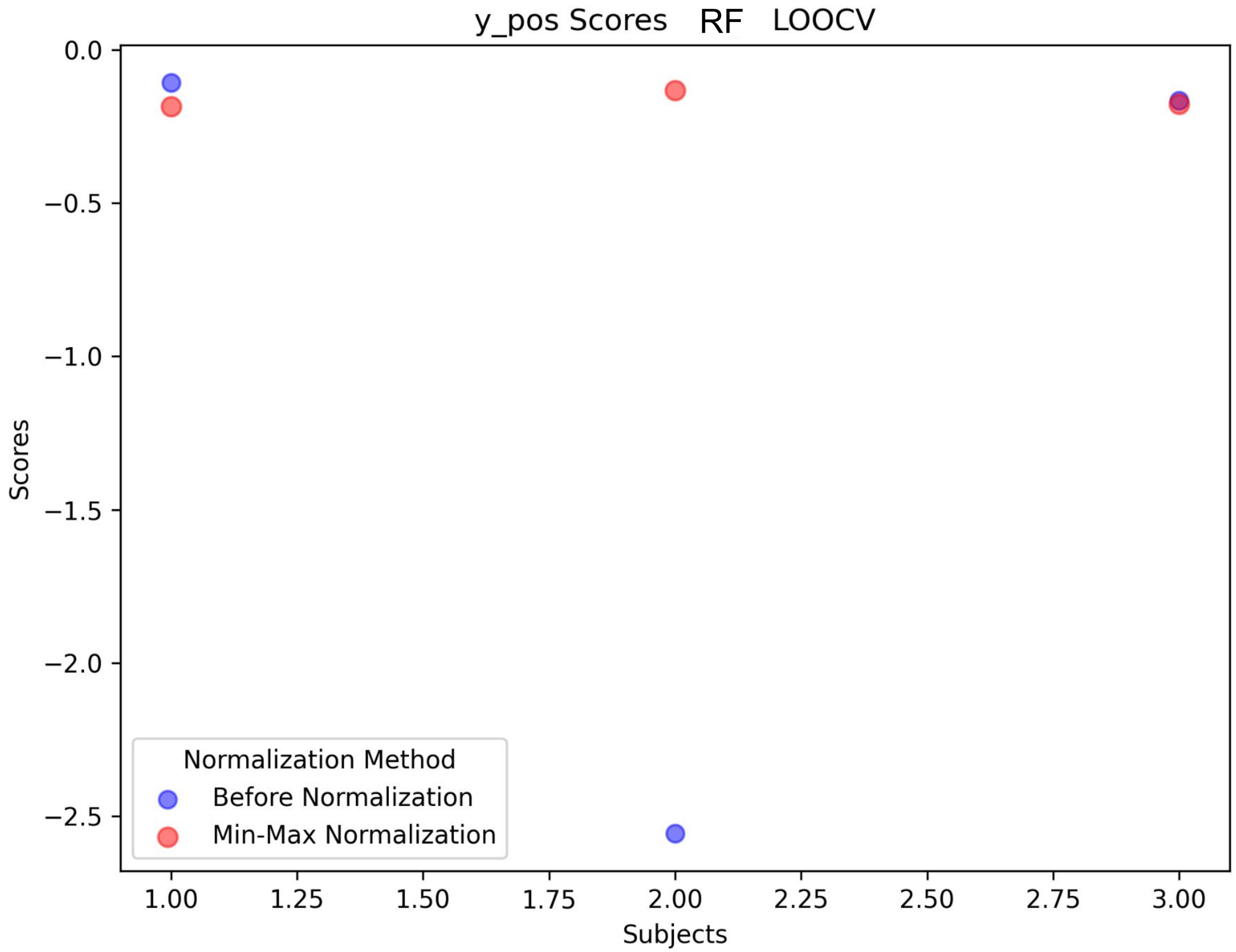
Results - Random Forest Regression

- Target: Position difference
- Leave one out cross validation
- 3 subjects, ~17000 data streams



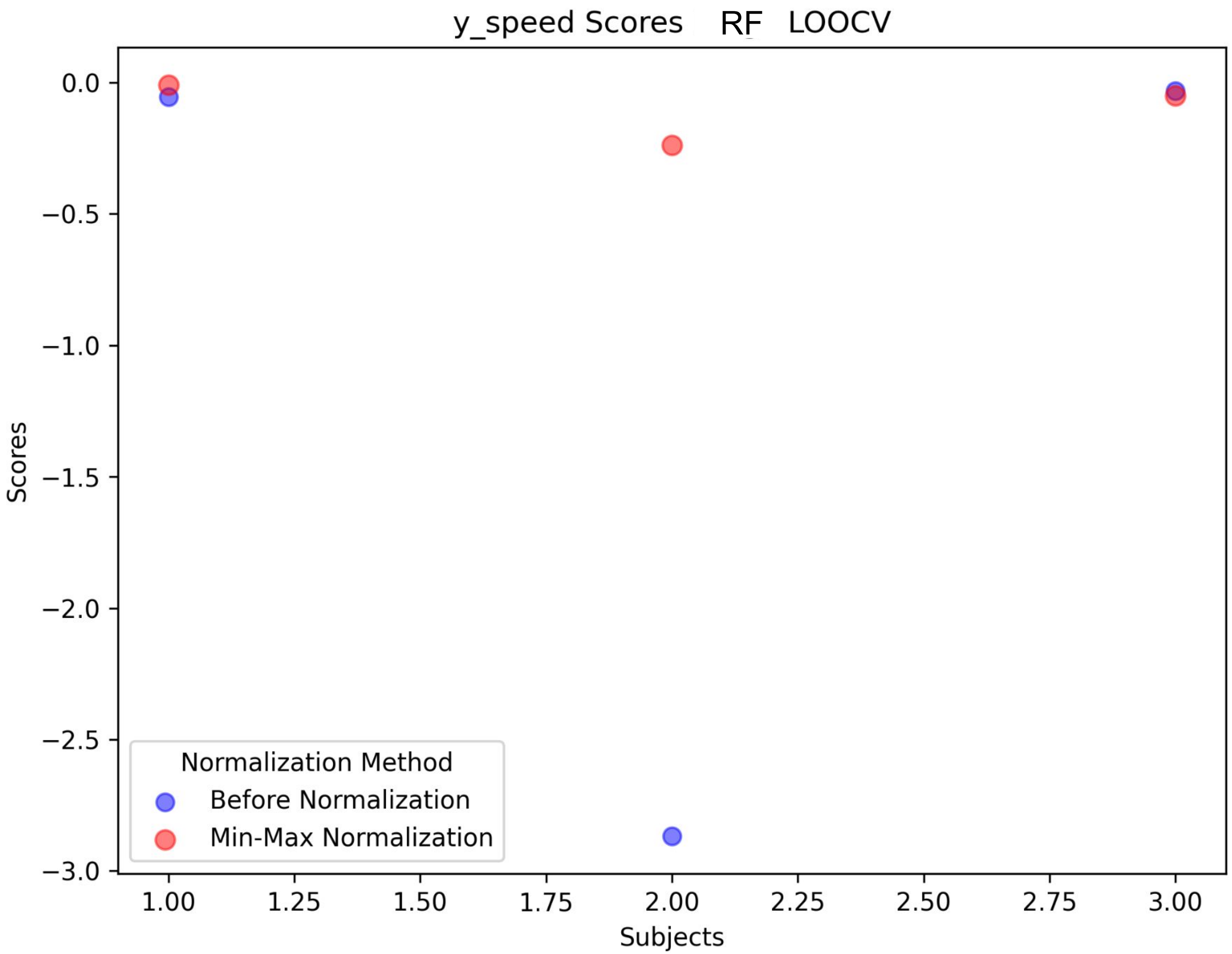
Results - Random Forest Regression

- Target: Stimulus position
- Leave one out cross validation
- 3 subjects, ~17000 data streams



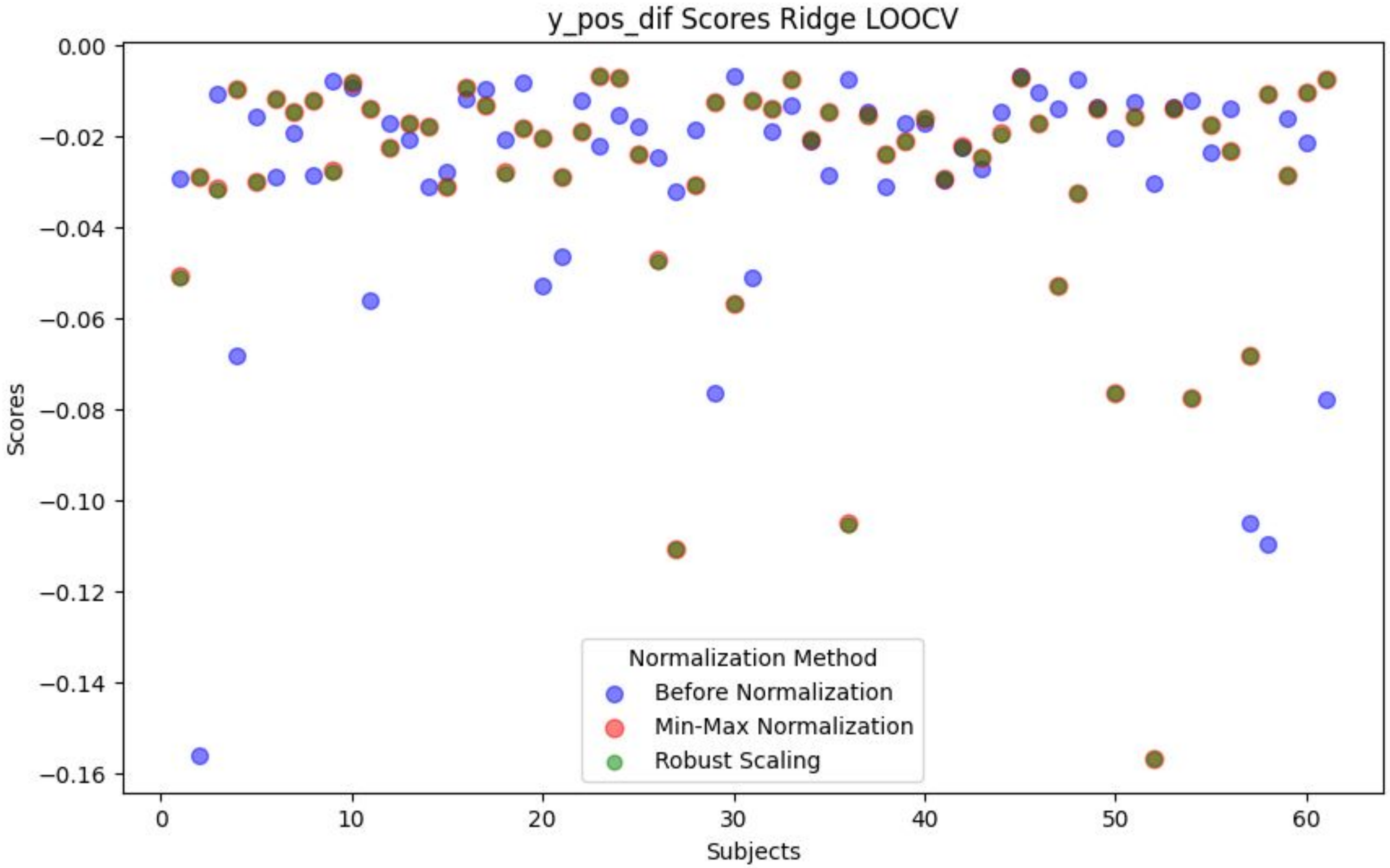
Results - Random Forest Regression

- Target: Stimulus speed
- Leave one out cross validation
- 3 subjects, ~17000 raw data streams



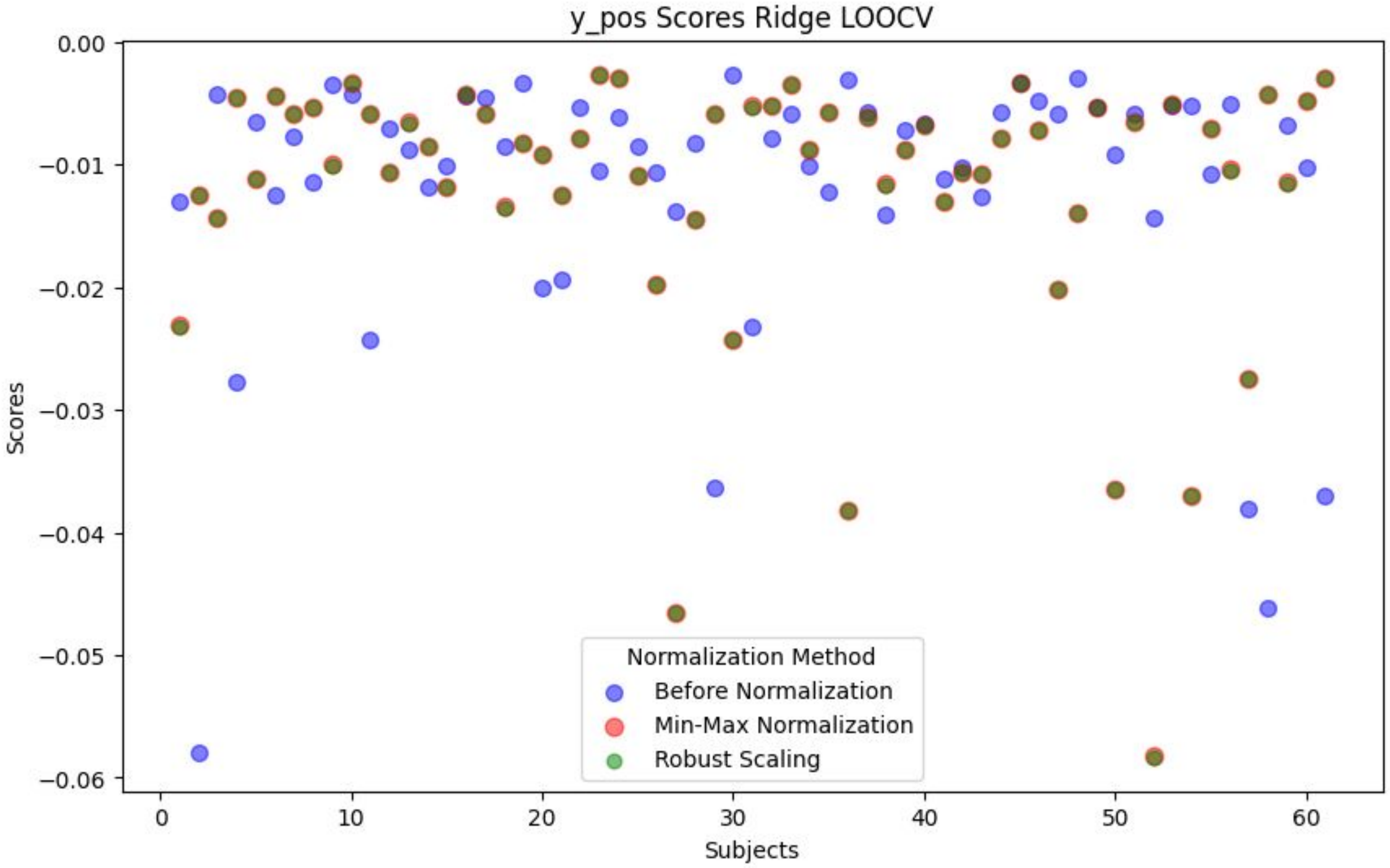
Results - Ridge Regression

- Target: Position Difference
- Leave one out cross validation
- 61 subjects, ~17000 data streams



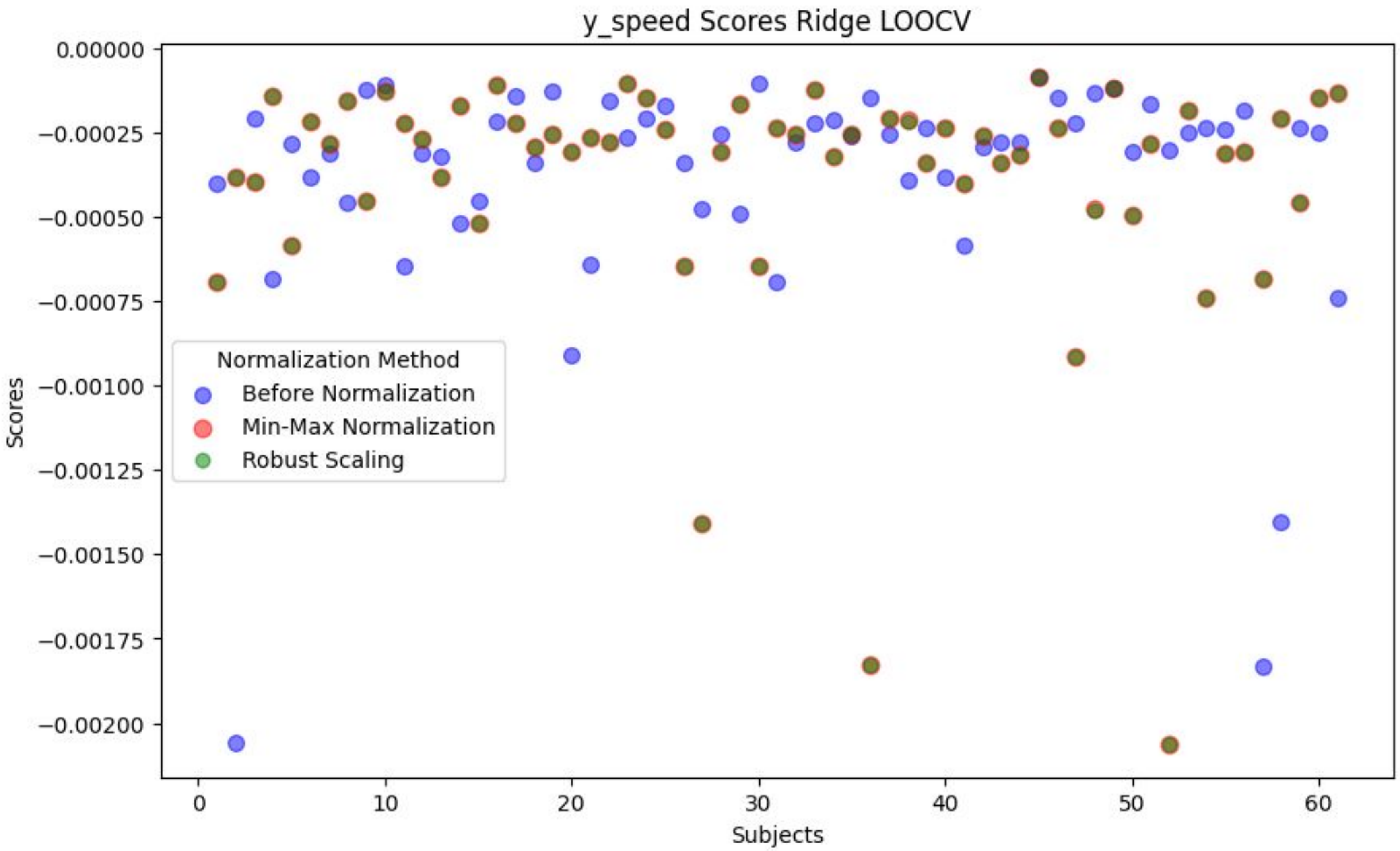
Results - Ridge Regression

- Target: Stimulus Position
- Leave one out cross validation
- 61 subjects, ~17000 data streams



Results - Ridge Regression

- Target: Stimulus Speed
- Leave one out cross validation
- 61 subjects, ~17000 data streams



Sprint	Task	Dates												
		Week 3 (9/9 - 9/15)	Week 4 (9/16 - 9/22)	Week 5 (9/23 - 9/29)	Week 6 (9/30 - 10/6)	Week 7 (10/7 - 10/13)	Week 8 (10/14 - 10/20)	Week 9 (10/21 - 10/27)	Week 10 (10/28 - 11/3)	Week 11 (11/4 - 11/10)	Week 12 (11/11 - 11/17)	Week 13 (11/18 - 11/24)	Week 14 (12/2 - 12/8)	Week 15 (12/9 - 12/15)
0	Replicate ridge regression algorithm													
	Preliminary LOO Cross Validation													Complete
	Run updated Mobi Code													In Process
1	R squared vs. # of Time points													Not Started
	Identify feature importance algorithms													
	Finalized HC and Pitch Deck													
	Replicate random forest regression													
2	Preprocessing - Normalization													
	Leave-one-out Cross Validation													
	Implement Time Series													
	Replicate feature importance algorithms													
3	Apply identified regression model													
	Apply feature importance													
	All 0.5 < R^2 < 1													

Final Presentation Slide Layout:

Comparing Ridge Regression code from Scratch vs Sklearn

Accuracy of the Model as the number of data points increases

Tuning of the alpha parameter

Attempts at various kind data normalization:

Min-max Normalization

Z-score Normalization

Robust Scaling

Pearson Correlation Analysis

Answering if Ridge Regression a viable algorithm for our Dataset

Comparing Ridge vs Random Forest

Questions ?